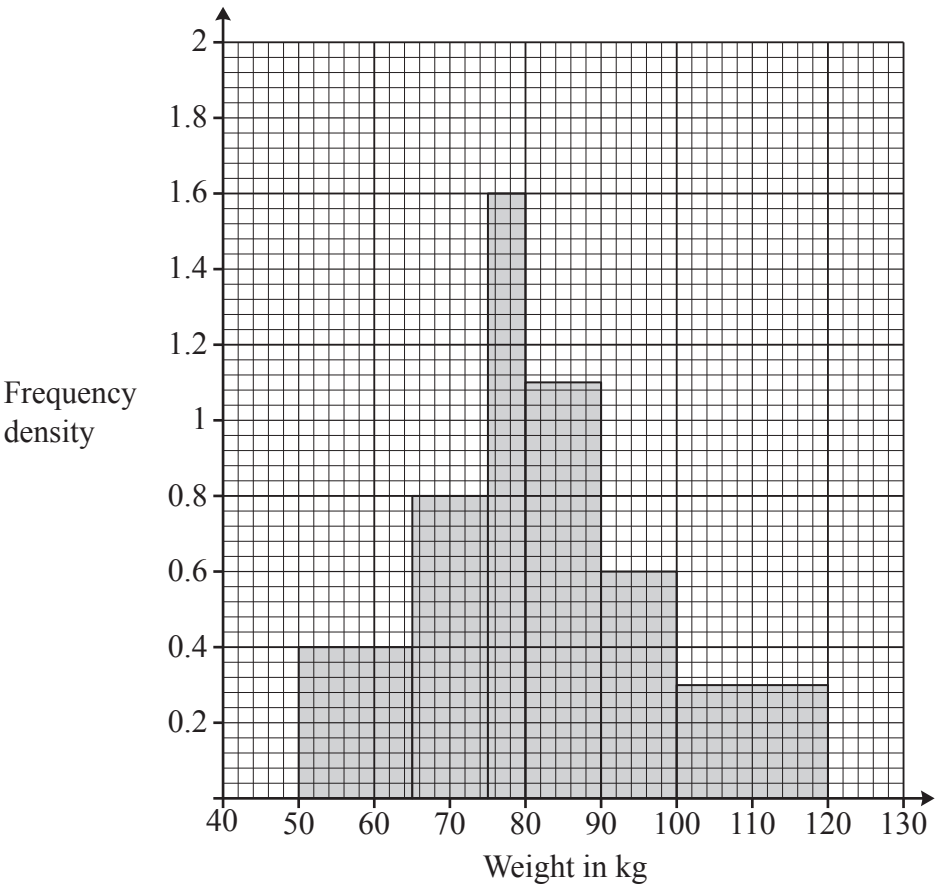
12 Doctors are investigating the weights of adult males registered at their surgery. One week they collect a sample by noting the weight in kilograms of all the adult males who have an appointment at their surgery.

(a) State the sampling method they use. **[1]**

(b) Explain why this method will **not** generate a simple random sample of all the adult males registered at their surgery. **[1]**

They represent the data using a histogram.



An incomplete frequency table for the data is shown below.

Weight in kg	50–	65–	75–	80–	90–	100–120
Frequency		8				

(c) Complete the copy of the frequency table in the Printed Answer Booklet. **[1]**

One of these patients is selected at random.

(d) Determine an estimate of the probability that he weighs either less than 60 kg or more than 110 kg. **[2]**

(e) Explain why your answer to part **(d)** is an estimate and not exact. **[1]**

12(a)

12(b)

12(c)

Weight in kg	50 –	65 –	75 –	80 –	90 –	100 – 120
Frequency		8				

12(d)

12(e)